

RESEARCH MANUSCRIPT

Sharp Gaussian Quadratic Concentration from Signed Spectral Moments

An optimal cubic-moment frontier, a four-moment certificate,
and applications to harmonic fluctuations and randomized estimation

Research draft with proofs and reproducible artifacts

8 October 2026

Abstract

Let $Q = g^\top A g - \text{tr } A$ for a standard real Gaussian vector and a nonzero symmetric matrix. Set $L = \|A\|_{\text{op}}$, $v = \text{tr } A^2$, and $s = \text{tr } A^3/(Lv)$. We determine the largest constant $c(s)$ for which $\mathbb{P}(Q \geq t) \leq \exp[-c(s) \min\{t^2/v, t/L\}]$ holds in every dimension at the specified value of s . The answer is explicit: it is the minimum of $1/4$ and a closed-form signed-Gamma rate at normalized deviation one. There is a transition at $s = -0.705589538603\dots$. In particular, $c(0) = 0.188714038110\dots$, compared with the established unrestricted constant $(1 - \log 2)/2 = 0.153426409720\dots$.

The proof retains the signed cubic spectral moment through a convex spectral-kernel envelope. A second refinement uses $\text{tr } A^4$ and a two-node Gauss–Radau rule. We give its exact moment-extremal property, closed-form Chernoff optimizer, equality cases, and nonnegative remainder. An unequal harmonic-mode contrast attains this envelope and has exactly computable rational tail probabilities, including a sharp asymptotic equivalent and its first correction. Gaussian trace estimates and diagonal-free subgaussian forms inherit the bounds; triangle-free random cuts provide a combinatorial application. The accompanying code reproduces the figures, exact certificates, symbolic identities, and numerical audits. Classical concentration and quadrature ingredients are explicitly credited; the claimed contribution is the moment-dependent sharpening and its optimality analysis.

Research status. The principal statements have complete proofs in this manuscript and were independently cross-checked during preparation. They have not undergone external peer review or Lean formalization. A targeted literature review found no matching statement of the complete cubic-moment constant frontier. This is a bounded novelty assessment, not a guarantee of priority. The unconditional constant and the general quadrature method are established results.

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1 Research target and relation to existing work

1.1 Why retain more than two norms?

Quadratic Gaussian fluctuations occur in sample covariance statistics, randomized trace estimates, likelihood comparisons, and harmonic Gibbs systems. A norm-only inequality treats a matrix with balanced positive and negative spectral contributions much like a positive projection. The variance can be identical although their upper tails are different. The cubic spectral moment is the first quantity that records this directional difference. The fourth moment records whether squared spectral mass is concentrated near the norm endpoint.

This article asks two precise questions. First, if the signed cubic spectral ratio is specified, what is the *best universal coefficient* in the familiar quadratic/linear concentration bound? Second, if one additional spectral moment is known, can the full exponential bound be improved without a numerical optimization over a large spectrum? The answers are affirmative and explicit. Neither conclusion changes the quadratic/linear asymptotic shape of Hanson–Wright concentration; the gain is a sharp information-dependent exponent.

1.2 Inspiration from the supplied repository

The supplied `openai/math` collection contains a family concerning Bloch’s law and its first lattice correction. In the inspected companion manuscript, the section on the explicit correction retains the lattice dispersion inside a heat-kernel integral before extracting an asymptotic term [1, 2]. We adopt the same useful methodological choice: retain an exact spectral kernel until the available spectral information has been used, then simplify. Here the kernel is an elementary Gaussian log-determinant, and its positivity makes the remainder controllable. No interacting-model result from that repository is an assumption of any theorem below. The resulting project is a self-contained strengthening in probability with direct applications in statistical physics.

1.3 The established baseline and the proposed contribution

The classical Laurent–Massart upper-tail inequality is

$$\mathbb{P}\{Q \geq 2\sqrt{vx} + 2Lx\} \leq e^{-x}, \quad x > 0, \quad (1.1)$$

initially stated for nonnegative weights [3]. For Gaussian forms one can retain the full Gamma envelope instead of its rational majorant. Its rate is

$$I_{\text{norm}}(t) = \frac{v}{2L^2} \left[\frac{tL}{v} - \log \left(1 + \frac{tL}{v} \right) \right]. \quad (1.2)$$

Dadush, Guzmán, and Olver give this functional form, and the coefficient $(1 - \log 2)/2$, in their work on diagonal-free subgaussian forms [4, Theorem 3.2 and Section 6]. Gallagher et al.’s Gaussian quadratic-form result specializes to the same optimizable envelope for general symmetric matrices [5]. Accordingly, neither (1.2) nor its universal coefficient is asserted to be new here.

Recent neighboring work includes Moshksar’s norm and Schatten-moment refinements [6] and Pang’s Gamma-difference comparison for absolute moments under Frobenius normalization [7]. The latter also retains a third centered moment, but addresses a different comparison class and optimization problem. Moment-based spectral quadrature itself is classical [8].

Component	Status in this article
Gaussian log-determinant and norm-only Gamma bound	Established tools, reproved to fix constants and conventions.
Signed-endpoint spectral envelope	An elementary convex-kernel specialization, related to Gamma-difference comparisons.
Complete optimal function $c(s)$	Principal proposed contribution: closed formula, transition, and sharpness at every real s .
Four-moment formula and optimizer	Explicit application of classical Radau extremality to a signed, variance-weighted log-MGF.
Exact rational harmonic example	A checkable equality case and sharpness certificate; beta/binomial and Stirling tools are classical.

The literature review and its limits are described further in Section 10. The goal is a theorem with a precise strengthening and a transparent proof, rather than an unsupported priority claim.

2 Normalization and the exact spectral kernel

Let A be a nonzero real symmetric $n \times n$ matrix, with eigenvalues $\lambda_1, \dots, \lambda_n$. Throughout,

$$\begin{aligned} Q &= g^\top A g - \text{tr } A, & g &\sim N(0, I_n), & L &= \|A\|_{\text{op}} > 0, \\ v &= \text{tr } A^2 > 0, & r &= \frac{v}{L^2}, & s &= \frac{\text{tr } A^3}{Lv}, & k &= \frac{\text{tr } A^4}{L^2 v}, & u &= \frac{tL}{v}. \end{aligned} \quad (2.1)$$

The letter k here is a normalized fourth spectral moment; a separate integer replication parameter will be explicitly introduced in Section 7. We have $1 \leq r \leq n$ and

$$-1 \leq s \leq 1, \quad s^2 \leq k \leq 1. \quad (2.2)$$

The parameter s is a *cubic spectral balance*, not the usual standardized skewness of Q . Indeed,

$$\text{Var}(Q) = 2v, \quad \mathbb{E}Q^3 = 8 \text{tr } A^3, \quad \frac{\mathbb{E}Q^3}{\text{Var}(Q)^{3/2}} = \frac{2\sqrt{2}s}{\sqrt{r}}. \quad (2.3)$$

Write $x_i = \lambda_i/L$ and introduce the probability measure

$$\nu = \frac{1}{r} \sum_{i=1}^n x_i^2 \delta_{x_i}. \quad (2.4)$$

Its support lies in $[-1, 1]$, and its first two moments are s, k . Zero eigenvalues contribute no mass. Cauchy–Schwarz and $x^2 \leq 1$ prove (2.2).

For $z \geq 0$ and $zx < 1$, set

$$g_z(x) = \frac{-\log(1 - zx) - zx}{2x^2}, \quad x \neq 0; \quad g_z(0) = \frac{z^2}{4}. \quad (2.5)$$

We reserve g without a subscript for the Gaussian vector; g_z is a scalar function. Gaussian integration and orthogonal invariance give

$$K_A(\theta) := \log \mathbb{E} e^{\theta Q} = \frac{1}{2} \sum_i [-\log(1 - 2\theta \lambda_i) - 2\theta \lambda_i] = r \int g_z(x) d\nu(x), \quad z = 2\theta L. \quad (2.6)$$

In particular this identity is valid for $0 \leq z < 1$, regardless of the signs of the eigenvalues.

Lemma 2.1 (Positive spectral derivatives). *For $0 < z < 1$ and $x \in [-1, 1]$,*

$$g_z(x) = \frac{z^2}{2} \int_0^1 \frac{a}{1 - zax} da, \quad (2.7)$$

and every derivative of g_z with respect to x is strictly positive. In particular, g_z is strictly increasing and strictly convex, and $g_z'''(x) > 0$.

Proof. Expand $(1 - zax)^{-1}$ in a uniformly convergent geometric series on the integration interval. The integral becomes $\sum_{j \geq 0} z^{j+2} x^j / [2(j+2)]$, which equals (2.5). Differentiating the integral j times in x gives

$$\frac{\partial^j g_z(x)}{\partial x^j} = \frac{z^{j+2} j!}{2} \int_0^1 \frac{a^{j+1}}{(1 - zax)^{j+1}} da > 0.$$

The denominator stays positive, including for negative x . $\square$

The variance weighting in (2.4) is decisive: moments $\text{tr } A^3$ and $\text{tr } A^4$ become the first and second moments of a probability measure, where sharp one-dimensional interpolation applies.

3 A signed cubic-moment exponential bound

Define, for $-1 \leq s \leq 1$ and $0 \leq z < 1$,

$$H_s(z) = -\frac{(1+s)\log(1-z) + (1-s)\log(1+z)}{4} - \frac{sz}{2}. \quad (3.1)$$

It is the normalized log-MGF of a centered difference of Gamma variables. This interpretation does not require integral Gamma shapes.

Theorem 3.1 (Signed-endpoint envelope). *For the quantities in (2.1),*

$$K_A\left(\frac{z}{2L}\right) \leq rH_s(z), \quad 0 \leq z < 1. \quad (3.2)$$

For $z > 0$, equality holds exactly when all nonzero eigenvalues have absolute value L . The right side is the simultaneous maximum over all probability measures on $[-1, 1]$ having first moment s .

If $-1 < s \leq 1$ and $u > 0$, put

$$z_s(u) = \frac{2u}{1 + \sqrt{1 + 4u(u+s)}}, \quad (3.3)$$

$$J_s(u) = \frac{uz_s(u)}{2} - H_s(z_s(u)). \quad (3.4)$$

Then

$$\mathbb{P}(Q \geq t) \leq \exp[-rJ_s(tL/v)]. \quad (3.5)$$

At $s = -1$, the same conclusion holds with

$$J_{-1}(u) = \begin{cases} [-u - \log(1-u)]/2, & 0 \leq u < 1, \\ +\infty, & u \geq 1. \end{cases} \quad (3.6)$$

Proof. The endpoint secant of the convex function g_z gives

$$g_z(x) \leq \frac{1+x}{2}g_z(1) + \frac{1-x}{2}g_z(-1).$$

Integrating and using $\int x d\nu = s$ proves (3.2). Strict convexity gives the equality condition. The measure $\frac{1+s}{2}\delta_1 + \frac{1-s}{2}\delta_{-1}$ attains the bound, proving moment extremality.

Chernoff's inequality gives the rate $\sup_{0 \leq z < 1} \{uz/2 - H_s(z)\}$. Direct differentiation yields

$$H'_s(z) = \frac{z(1+sz)}{2(1-z^2)}, \quad H''_s(z) = \frac{1+2sz+z^2}{2(1-z^2)^2} > 0. \quad (3.7)$$

For $s > -1$, the tilted mean $2H'_s(z)$ increases from zero to infinity. The unique optimizer solves

$$u = \frac{z(1+sz)}{1-z^2}, \quad (3.8)$$

whose root in $(0, 1)$ is (3.3). This proves (3.5).

If $s = -1$, the measure ν is concentrated at -1 , so A is a negative scalar multiple of a projection. The log-MGF has no positive singularity: $H_{-1}(z) = [z - \log(1+z)]/2$ for all $z \geq 0$. Optimization gives $z = u/(1-u)$ for $u < 1$ and the rate (3.6). At or above $u = 1$ the event has probability zero. $\square$

Remark 3.2 (Comparison with the norm-only bound). For $z > 0$,

$$\frac{\partial H_s(z)}{\partial s} = \frac{1}{4} \log \frac{1+z}{1-z} - \frac{z}{2} > 0.$$

Consequently $J_s(u)$ decreases with s , strictly for $u > 0$ and $-1 < s < 1$. At $s = 1$, $J_1(u) = [u - \log(1+u)]/2$, recovering (1.2). Every value $s < 1$ therefore yields a strict improvement of that upper-tail envelope, except for deterministic support endpoints.

The lower tail follows by applying the theorem to $-A$. In particular,

$$\mathbb{P}(|Q| \geq t) \leq \min\{1, e^{-rJ_s(u)} + e^{-rJ_{-s}(u)}\}. \quad (3.9)$$

Negative s improves the upper tail; the opposite tail sees $-s$. This directional distinction is necessary.

4 The exact optimal coefficient frontier

Theorem 4.1 (Optimal coefficient at fixed cubic spectral balance). *For $-1 < s \leq 1$, define*

$$c(s) = \min\{1/4, J_s(1)\}, \quad c(-1) = 1/4. \quad (4.1)$$

For every nonzero real symmetric A and every $t > 0$,

$$\mathbb{P}(Q \geq t) \leq \exp \left[-c(s) \min \left\{ \frac{t^2}{v}, \frac{t}{L} \right\} \right]. \quad (4.2)$$

For each specified real $s \in [-1, 1]$, no larger coefficient works for all dimensions and all matrices having exactly that value of s . The same coefficient is optimal for the larger class with cubic ratio at most s .

The result concerns the largest constant in a specified functional form with prefactor one. It does not assert that its exponential right side is the exact finite-dimensional tail probability.

4.1 Finding the global minimum

Proof of the inequality and coefficient formula. The required coefficient from Theorem 3.1 is

$$\inf_{u>0} \frac{J_s(u)}{\min\{u^2, u\}}.$$

Since J_s is convex and $J_s(0) = 0$, the ratio $J_s(u)/u$ is nondecreasing. Thus $u \geq 1$ contributes its minimum at $u = 1$.

For $0 < u \leq 1$, let $z = z_s(u)$. Legendre differentiation gives $J'_s(u) = z/2$. Therefore the sign of the derivative of $J_s(u)/u^2$ is the sign of

$$uJ'_s(u) - 2J_s(u) = 2H_s(z) - zH'_s(z) =: D_s(z). \quad (4.3)$$

A direct differentiation, with no inequality, gives

$$D'_s(z) = -\frac{z^2\{s(1+z^2) + 2z\}}{2(1-z^2)^2}. \quad (4.4)$$

For $s \geq 0$, this is negative, so $J_s(u)/u^2$ decreases. For $-1 < s < 0$, the bracket in braces is strictly increasing on $(0, 1)$: its derivative is $2 + 2sz > 0$. It starts negative and ends positive. Thus D_s first increases and then decreases. Since $D_s(0) = 0$, its sign can change only from positive to negative. The ratio $J_s(u)/u^2$ can have an interior maximum, but no interior minimum. Its infimum on $(0, 1]$ is consequently one of its endpoint values.

The expansion $H_s(z) = z^2/4 + sz^3/6 + O(z^4)$ implies

$$J_s(u) = \frac{u^2}{4} - \frac{su^3}{6} + O(u^4). \quad (4.5)$$

The endpoint values are $1/4$ and $J_s(1)$, proving (4.1). For $s = -1$, the elementary inequality $z - \log(1+z) \leq z^2/2$ for $z \geq 0$ gives $\mathbb{P}(Q \geq t) \leq e^{-t^2/(4v)}$, which implies (4.2) with $c(-1) = 1/4$. $\square$

4.2 Transition and special values

The function $s \mapsto J_s(1)$ is continuous and strictly decreasing on $(-1, 1]$. At its left endpoint it tends to $(\log 2)/2 > 1/4$, while $J_1(1) = (1 - \log 2)/2 < 1/4$. There is a unique transition

$$J_{s_{\text{crit}}}(1) = 1/4, \quad s_{\text{crit}} = -0.705589538603\dots \quad (4.6)$$

Thus $c(s) = 1/4$ for $-1 \leq s \leq s_{\text{crit}}$, and $c(s) = J_s(1)$ above that interval. If $\varphi = (1 + \sqrt{5})/2$, then

$$\begin{aligned} c(1) &= \frac{1 - \log 2}{2} = 0.153426409720\dots, \\ c(0) &= \frac{\sqrt{5} - 1 - \log \varphi}{4} = 0.188714038110\dots \end{aligned} \quad (4.7)$$

The second coefficient is approximately 23.00% larger. This is a gain in the exponent, not a uniform 23% statement about the actual tail probabilities.

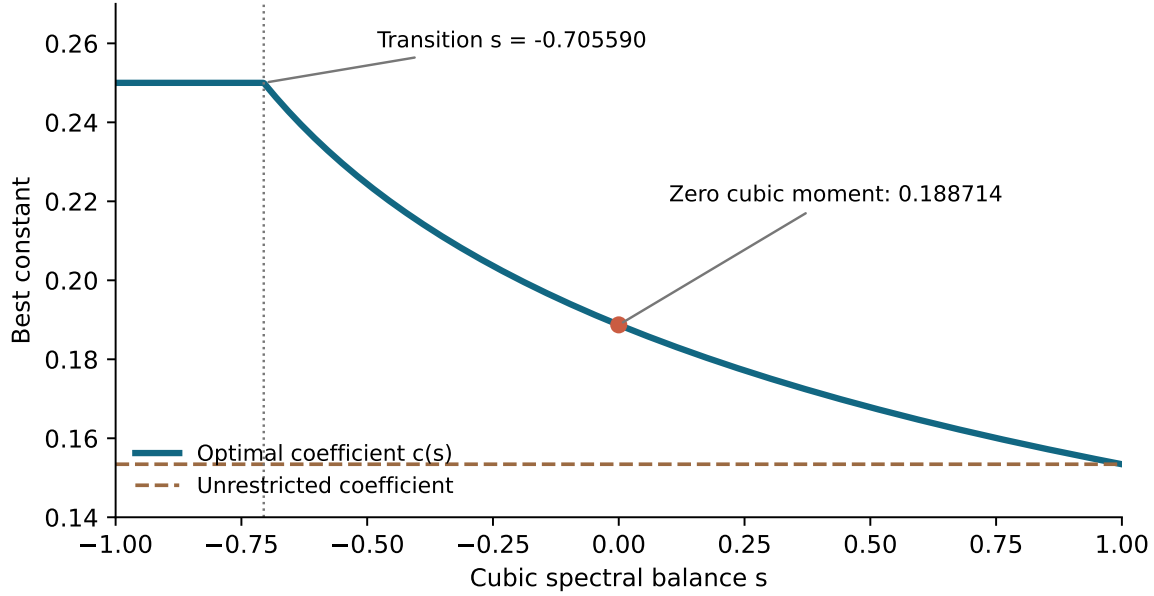


Figure 1: The proved optimal coefficient at each cubic spectral balance. The plateau comes from arbitrarily small normalized deviations; the decreasing branch comes from normalized deviation $u = 1$.

4.3 Sharpness for every real parameter

We supply the exact-parameter argument because rounding positive and negative multiplicities alone does not prove sharpness when s is irrational.

Lemma 4.2 (Exact cubic balance and limiting extremizers). *Fix $s \in (-1, 1)$. For all sufficiently large integers n , there are symmetric $n \times n$ matrices A_n with $\|A_n\|_{\text{op}} = 1$, exact cubic ratio s , and $v_n/n \rightarrow 1$ such that for every fixed $u > 0$,*

$$\lim_{n \rightarrow \infty} \frac{1}{v_n} \log \mathbb{P}\{Q_n \geq v_n u\} = -J_s(u). \quad (4.8)$$

Proof. Put $p_n = \lfloor (1+s)n/2 \rfloor$ and $q_n = n - p_n$. For sufficiently large n both are positive. The function

$$F_n(a) = \frac{p_n - q_n a^3}{p_n + q_n a^2}, \quad 0 \leq a \leq 1,$$

satisfies $F_n(0) = 1$, $F_n(1) \leq s$, and

$$F'_n(a) = -\frac{q_n a(2p_n + 3p_n a + q_n a^3)}{(p_n + q_n a^2)^2} < 0 \quad (a > 0).$$

There is a unique $a_n \in (0, 1]$ with $F_n(a_n) = s$. Since $F_n(1) = s + O(1/n)$ and the limiting derivative near one is nonzero, $a_n = 1 + O_s(1/n)$. Define

$$A_n = \text{diag}(1^{[p_n]}, (-a_n)^{[q_n]}).$$

Then $v_n = p_n + q_n a_n^2$, $v_n/n \rightarrow 1$, and the cubic ratio is exactly s . The normalized log-MGFs $K_{A_n}(\theta)/v_n$, together with their first two derivatives, converge locally uniformly for $0 \leq \theta < 1/2$ to $H_s(2\theta)$.

The upper bound in (4.8) follows directly from Theorem 3.1. For a lower bound fix $\delta > 0$ and tilt the original probability measure at $\theta = z_s(u + \delta)/2$:

$$\frac{d\mathbb{P}_\theta}{d\mathbb{P}} = \exp\{\theta Q_n - K_{A_n}(\theta)\}.$$

Under this measure, $\mathbb{E}_\theta Q_n/v_n \rightarrow u + \delta$ and $\text{Var}_\theta(Q_n)/v_n^2 = O(1/v_n)$. Chebyshev's inequality therefore shows that the tilted probability of

$$v_n(u + \delta/2) \leq Q_n \leq v_n(u + 3\delta/2)$$

tends to one. Returning to the original measure gives

$$\liminf_n \frac{1}{v_n} \log \mathbb{P}(Q_n \geq v_n u) \geq -J_s(u + \delta) - \frac{\delta}{2} \frac{z_s(u + \delta)}{2}.$$

Let $\delta \downarrow 0$ to obtain (4.8). □

Completion of Theorem 4.1. If $c(s) = J_s(1) < 1/4$, use Lemma 4.2 at $u = 1$. If $c(s) = 1/4$, first choose a small fixed $u > 0$ so that $J_s(u)/u^2$ is arbitrarily close to $1/4$, then let n grow. In both cases any larger coefficient contradicts (4.8). For $s = 1$ and $s = -1$, use $A_n = I_n$ and $A_n = -I_n$ respectively; the same tilting argument works, with $u < 1$ for the negative case. Finally $c(s)$ is nonincreasing, so it is valid for the class with cubic ratio at most s . The exact- s subclass already proves optimality. □

For a two-sided bound in the same simple form one obtains

$$\mathbb{P}(|Q| \geq t) \leq 2 \exp[-c(|s|) \min\{t^2/v, t/L\}]. \quad (4.9)$$

The asymmetric sum (3.9) usually retains more information.

5 An explicit four-moment refinement

For feasible (s, k) with $s < 1$, define

$$a = \frac{s - k}{1 - s}, \quad w = \frac{k - s^2}{1 - 2s + k}. \quad (5.1)$$

Here $-1 \leq a \leq s < 1$, $0 \leq w < 1$, and

$$\nu_R = (1 - w)\delta_a + w\delta_1 \quad (5.2)$$

has moments $1, s, k$. Let

$$B_R(s, k; z) = (1 - w)g_z(a) + wg_z(1). \quad (5.3)$$

When $s = 1$, feasibility forces $k = 1$ and we define $a = 1, w = 1$ and $B_R(1, 1; z) = g_z(1)$.

Theorem 5.1 (Four-moment extremal log-MGF). *For $0 \leq z < 1$,*

$$\frac{1}{r} K_A\left(\frac{z}{2L}\right) \leq B_R(s, k; z) \leq H_s(z) \leq H_1(z). \quad (5.4)$$

For every $0 < z < 1$, $B_R(s, k; z)$ is the exact maximum of $\int g_z d\eta$ among all probability measures η on $[-1, 1]$ with first two moments s, k . The same extremizing measure works for all such z . For $0 < z < 1$, equality in the first inequality holds precisely when every nonzero eigenvalue of A belongs to $\{aL, L\}$.

Proof. For $s < 1$, the inequalities $s^2 \leq k \leq 1$ imply $a \leq s$ and $a \geq -1$. Direct substitution verifies the first two moments of (5.2). Let P_z be the quadratic Hermite interpolant to g_z that matches its value and first derivative at a and its value at 1. Explicitly,

$$P_z(x) = g_z(a) + g'_z(a)(x - a) + \frac{g_z(1) - g_z(a) - g'_z(a)(1 - a)}{(1 - a)^2}(x - a)^2.$$

The interpolation remainder is

$$g_z(x) - P_z(x) = \frac{g_z'''(\xi_x)}{6}(x - a)^2(x - 1) \leq 0.$$

It is strictly negative away from the interpolation nodes. Since P_z has degree two, integrating it against any admissible measure gives the same value as integrating against ν_R . At that measure interpolation is exact. This proves extremality and the equality statement. The endpoint secant gives $B_R \leq H_s$, and the last comparison was proved above. The case $s = 1$ is immediate. $\square$

This proof is the elementary two-node Gauss–Radau argument, applied to the variance-weighted spectral measure; classical general versions are described in [8]. For $s^2 < k < 1$, the improvement $B_R(s, k; z) < H_s(z)$ is strict for every $0 < z < 1$.

5.1 Closed-form optimization

If $s < 1$ and $k > s^2$, then $w > 0$, so the upper endpoint creates a singularity at $z = 1$. Define

$$z_R(s, k; u) = \frac{2u}{1 + u(1 + a) + \sqrt{[1 + u(2s - 1 - a)]^2 + 4u^2(k - s^2)}}. \quad (5.5)$$

Set

$$J_R(s, k; u) = \frac{uz_R}{2} - B_R(s, k; z_R). \quad (5.6)$$

Corollary 5.2 (Explicit four-moment tail bound). *For nondegenerate data as above,*

$$\boxed{\mathbb{P}(Q \geq t) \leq \exp[-rJ_R(s, k; tL/v)]}. \quad (5.7)$$

The exponent satisfies $J_R(s, k; u) \geq J_s(u) \geq J_1(u)$. Computing it from the four scalar inputs requires a square root and at most two logarithms, with no scalar root search.

Proof. Differentiate (2.5) with respect to z :

$$\partial_z g_z(x) = \frac{z}{2(1 - zx)}, \quad \partial_z^2 g_z(x) = \frac{1}{2(1 - zx)^2}.$$

The optimizer is unique and satisfies

$$u = z \left(\frac{1 - w}{1 - za} + \frac{w}{1 - z} \right). \quad (5.8)$$

Clearing denominators produces the quadratic

$$[ua + 1 + a - s]z^2 - [1 + u(1 + a)]z + u = 0.$$

Its unique root in $(0, 1)$, written to avoid subtracting nearby quantities, is (5.5). Applying Chernoff proves the result. The ordering of exponents follows from the pointwise ordering of log-MGFs. $\square$

Degenerate data. If $k = s^2$, then $w = 0$ and the spectral measure is δ_s . The artificial endpoint 1 carries no mass and must not restrict the optimization domain. For $s \neq 0$ the rate is

$$J_R(s, s^2; u) = \frac{us - \log(1 + us)}{2s^2}, \quad 1 + us > 0. \quad (5.9)$$

For $s < 0$ it is $+\infty$ at $u \geq -1/s$. At $s = 0, k = 0$ the relaxed envelope is Gaussian and its rate is $u^2/4$. This last pair cannot arise from a nonzero finite matrix with $L = \|A\|_{\text{op}}$. More generally, with that exact norm the boundary $k = s^2$ is attainable only at $s = 1$ or $s = -1$: a one-point nonzero spectral support must itself attain the norm. All other boundary formulas describe the relaxed moment problem or a larger supplied norm bound. The endpoint $s = 1, k = 1$ recovers J_1 .

5.2 What moment optimality means

The extremal assertion in Theorem 5.1 is exact for a *probability-measure moment problem*. An atom of mass ω at a nonzero normalized eigenvalue x would require multiplicity $r\omega/x^2$ in a finite matrix. This need not be an integer. An atom at zero with positive variance weight is impossible for a finite spectrum, though it can be approached by many small eigenvalues. Therefore moment extremality alone does not prove finite-matrix minimax optimality at every prescribed triple (r, s, k) .

Each atom nevertheless has a precise probabilistic realization:

$$x(\chi_{r\omega/x^2}^2 - r\omega/x^2), \quad x \neq 0,$$

with Gamma-defined, possibly nonintegral degrees of freedom. A zero atom contributes an independent Gaussian of variance $2r\omega$. The Radau log-MGF is the sum of these centered log-MGFs after scaling by L . When multiplicities are integral and nodes nonzero, it is an actual Gaussian quadratic form. For L also to be its exact operator norm, some positive spectral mass must have $|x| = 1$. Section 7 gives such an example with an exact tail calculation.

5.3 A coefficient for the four-moment envelope

There is also a simple quadratic/linear coefficient:

$$C_R(s, k) = \min\{1/4, J_R(s, k; 1)\} \geq c(s). \quad (5.10)$$

It is valid in (4.2) in place of $c(s)$ and is the optimal coefficient extracted from this log-MGF envelope. This last wording is distinct from the finite-matrix minimax question just discussed.

Lemma 5.3 (An endpoint rule for these rate functions). *Let $F(0) = F'(0) = 0$, $F''(0) = 1/2$, $F''' > 0$, and $F'''' \geq 0$ on its nonnegative log-MGF domain. Suppose $2F'$ covers all positive values and let $J(u) = \sup_z \{uz/2 - F(z)\}$. Then*

$$\inf_{u>0} \frac{J(u)}{\min\{u^2, u\}} = \min\{1/4, J(1)\}.$$

Proof. Convexity again handles $u \geq 1$. In the interior, $J'''(u) = -F''''(z)/(8F''(z)^3)$. Since F'''' is nondecreasing, J''' can change sign only from positive to negative. For $G(u) = uJ'(u) - 2J(u)$ we have $G(0) = G'(0) = 0$ and $G''(u) = uJ'''(u)$. It follows successively that G' and G either have one sign throughout or change sign only from positive to negative. Thus $J(u)/u^2$, whose derivative has sign G , cannot take an interior value below both endpoint values. Its limiting value at zero is $1/4$. $\square$

For $F = B_R$ the assumptions hold when $w > 0$, because

$$F''''(z) = 3 \left[\frac{(1-w)a^2}{(1-za)^4} + \frac{w}{(1-z)^4} \right] \geq 0.$$

The single-node boundary cases follow directly from (5.9). This proves (5.10).

6 Explicit deficits and controlled remainders

6.1 A simple fourth-moment correction to the cubic bound

The Radau formula is inexpensive, but an additive correction reveals how the fourth moment improves a cubic-only estimate. Define

$$D(z) = \frac{1}{2} \int_0^z \frac{y^3}{(1-y)(1+y)^2} dy \tag{6.1}$$

$$= -\frac{z}{2} - \frac{1}{8} \log(1-z) + \frac{5}{8} \log(1+z) + \frac{1}{4(1+z)} - \frac{1}{4}. \tag{6.2}$$

It is positive for $0 < z < 1$ and $D(z) = z^4/8 + O(z^5)$.

Proposition 6.1 (Spectral spread gives a quantitative gain). *For $0 \leq z < 1$,*

$$\frac{1}{r} K_A \left(\frac{z}{2L} \right) \leq H_s(z) - (1-k)D(z). \tag{6.3}$$

For $s > -1$, this implies the explicit bound

$$\mathbb{P}(Q \geq t) \leq \exp\{-r[J_s(u) + (1-k)D(z_s(u))]\}. \tag{6.4}$$

The Radau envelope is at least as strong as the right side of (6.3) at every z .

Proof. For $0 \leq h < 1$, the endpoint secant of $(1-hx)^{-1}$ satisfies the exact identity

$$\frac{1+x}{2(1-h)} + \frac{1-x}{2(1+h)} - \frac{1}{1-hx} = \frac{h^2(1-x^2)}{(1-h^2)(1-hx)}.$$

Integrating this with $h = za$ against $z^2 a da/2$ gives the secant defect of g_z . Since $1-za \leq 1+za$, this defect is at least $(1-x^2)D(z)$. Integration against ν proves (6.3). Evaluation at the old saddle proves (6.4). Since (6.3) holds for every measure with these moments and Radau attains their supremum, it also bounds B_R . $\square$

The coefficient $D(z)$ is the largest pointwise coefficient in this secant-defect comparison: the ratio of the defect to $1-x^2$ increases with x and tends to $D(z)$ as $x \downarrow -1$. Useful elementary bounds are

$$D(z) \geq \frac{z^4}{8(1+z)}, \quad D(z) \geq \frac{27z^4}{256}. \tag{6.5}$$

Indeed $(1-y)(1+y)^2 \leq 1+y \leq 1+z$, and its maximum on $[0, 1]$ is $32/27$.

6.2 The first information lost by Radau

Let a be the interior node from (5.1), and set

$$\rho_5 = \int (1-x)(x-a)^2 d\nu(x) \geq 0. \quad (6.6)$$

The rational Hermite remainder, followed by (2.7), gives the exact identity

$$B_R(s, k; z) - \frac{1}{r} K_A \left(\frac{z}{2L} \right) = \frac{z^5}{2} \int_0^1 \int_0^1 \frac{b^4 (1-x)(x-a)^2}{(1-zbx)(1-zba)^2(1-zb)} db d\nu(x). \quad (6.7)$$

Here b is only an integration variable. To verify the identity, the Hermite interpolant of $(1-hx)^{-1}$ at the double node a and single node 1 has excess

$$\frac{h^3(1-x)(x-a)^2}{(1-hx)(1-ha)^2(1-h)}.$$

It agrees at the two nodes and has the required derivative; clearing denominators verifies the formula directly. Integrating it proves (6.7).

Consequently Radau preserves the exact cumulants through fourth order, and

$$B_R - \frac{K_A}{r} = \frac{\rho_5 z^5}{10} + O(z^6). \quad (6.8)$$

Uniformly for $0 \leq z < 1$,

$$\frac{\rho_5 z^5}{10(1+z)^4} \leq B_R - \frac{K_A}{r} \leq \frac{\rho_5 z^5}{10(1-z)^4}. \quad (6.9)$$

In unnormalized notation, if $p_j = \text{tr } A^j$ and $c = aL$, the leading physical residual is

$$R_5 = Lp_4 - p_5 - c^2(Lp_2 - p_3) = \sum_i \lambda_i^2 (L - \lambda_i)(\lambda_i - c)^2 \geq 0, \quad (6.10)$$

and the log-MGF difference is $(16/5)R_5\theta^5 + O(\theta^6)$. These expressions identify exactly what a fifth spectral moment could improve. Their nonnegativity remains valid for indefinite matrices.

7 An exact harmonic example with rational tail certificates

Consider a matrix with m eigenvalues 1 and $8m$ eigenvalues $-1/2$:

$$A = \text{diag}(1^{[m]}, (-1/2)^{[8m]}). \quad (7.1)$$

Its parameters are

$$L = 1, \quad v = r = 3m, \quad s = 0, \quad k = \frac{1}{2}, \quad \nu = \frac{1}{3}\delta_1 + \frac{2}{3}\delta_{-1/2}.$$

This is exactly the Radau measure for these moments. At normalized deviation $u = 1$, the optimizer is $z_R = 2/3$ and

$$J_* := J_R(0, 1/2; 1) = \frac{4}{3} \log \frac{4}{3} - \frac{1}{6} \log 3 = 0.200474048491 \dots \quad (7.2)$$

The comparison is therefore exact at the log-MGF level:

Information used	Exponent divided by v , at $t = v$
Norm and variance	0.153426409720...
Also the signed cubic moment	0.188714038110...
Also the fourth moment	0.200474048491...
Full spectrum	0.200474048491...

The fourth moment gains another 6.23% in this exponent beyond the cubic-moment bound.

7.1 An exact probability, without simulation

For this subsection only, put $m = 2\ell$ with positive integer ℓ . Then $v = 6\ell$, and

$$Q \stackrel{d}{=} X - \frac{1}{2}Y + 6\ell, \quad X \sim \chi_{2\ell}^2, \quad Y \sim \chi_{16\ell}^2$$

with independent X, Y . At $Q \geq v$, the event is $X \geq Y/2$. The change of variables $B = X/(X + Y)$ in the joint Gamma density shows that B has beta density proportional to $b^{\ell-1}(1-b)^{8\ell-1}$. For positive integer beta parameters, differentiation of the binomial tail and checking its endpoint value gives

$$P_\ell := \mathbb{P}(Q \geq v) = 3^{-(9\ell-1)} \sum_{j=0}^{\ell-1} \binom{9\ell-1}{j} 2^{9\ell-1-j}. \quad (7.3)$$

Thus every value is an exact rational number; $P_1 = 256/6561$. The code stores the integer numerator and denominator, not a Monte Carlo estimate.

Proposition 7.1 (Sharp equivalent and first correction). *For the probabilities in (7.3),*

$$P_\ell = \frac{e^{-6\ell J_*}}{3\sqrt{\pi\ell}} \left(1 - \frac{505}{864\ell} + O(\ell^{-2}) \right). \quad (7.4)$$

In particular $-\log P_\ell/(6\ell) \rightarrow J_$, so the four-moment exponent is asymptotically attained by actual finite matrices.*

Proof. Let T_ℓ be the last summand of (7.3), including the common normalization. A binomial identity gives

$$T_\ell = \frac{1}{3} \binom{9\ell}{\ell} \frac{2^{8\ell}}{3^{9\ell}}.$$

Stirling's formula with its usual controlled remainder [11] yields

$$T_\ell = \frac{e^{-6\ell J_*}}{4\sqrt{\pi\ell}} \left(1 - \frac{73}{864\ell} + O(\ell^{-2}) \right). \quad (7.5)$$

Moving backward by h summands multiplies T_ℓ by

$$R_{\ell,h} = \prod_{i=1}^h \frac{2(\ell-i)}{8\ell+i}, \quad 0 \leq h \leq \ell-1.$$

Each factor is at most $1/4$, and for fixed h , $R_{\ell,h} \rightarrow 4^{-h}$. This already proves the leading equivalent by dominated convergence, since $\sum_{h \geq 0} 4^{-h} = 4/3$.

For the correction, expand the logarithm of the product:

$$R_{\ell,h} = 4^{-h} \left(1 - \frac{9h(h+1)}{16\ell} + O\left(\frac{(1+h)^4}{\ell^2}\right) \right), \quad h \leq \ell^{1/8}.$$

The remainder is uniform on this range by the Taylor remainder for $\log(1+x)$. The sum beyond this range is bounded by a geometric tail, hence smaller than any inverse power of ℓ . Since

$$\frac{\sum_{h \geq 0} h(h+1)4^{-h}}{\sum_{h \geq 0} 4^{-h}} = \frac{8}{9},$$

we obtain $\sum_h R_{\ell,h} = (4/3)(1 - 1/(2\ell) + O(\ell^{-2}))$. Multiplication by (7.5) proves (7.4). $\square$

There is also an exact finite sandwich:

$$T_\ell \leq P_\ell \leq \frac{T_\ell}{1 - 2(\ell - 1)/(8\ell + 1)}. \quad (7.6)$$

It follows because each successive backward ratio is at most the first. The included standard-library script verifies this sandwich in exact rational arithmetic.

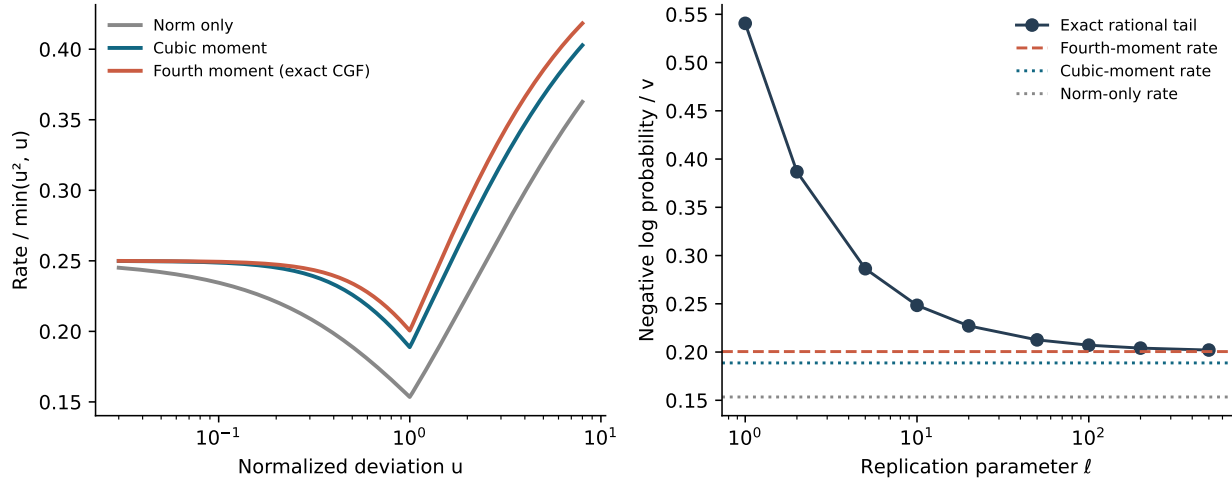


Figure 2: The example (7.1). Left: normalized exponents from two, three, and four moments. Right: exact rational tails approach the fourth-moment exponent. The horizontal variable ℓ is the integer replication parameter of this section.

ℓ	Dimension	$-\log P_\ell/(6\ell)$	$3\sqrt{\pi\ell} e^{6\ell J_*} P_\ell$
1	18	0.5406201441	0.6908038618
5	90	0.2863652587	0.9038967705
20	360	0.2271143501	0.9723953442
100	1800	0.2071063054	0.9942249100
500	9000	0.2020671979	0.9988338567

The first ratio tends to J_* and the second to one. These finite data check the prediction but do not replace its proof.

8 Applications

8.1 Harmonic Gibbs fluctuations and free energy

Let K be a real symmetric positive-definite stiffness matrix and $\beta > 0$ the inverse temperature. The Gibbs density for

$$\mathcal{H}(x) = \frac{1}{2}x^\top Kx$$

is Gaussian with covariance $\beta^{-1}K^{-1}$. For any symmetric matrix B , define $\mathcal{O}_B(x) = x^\top Bx/2$. Whitening gives the exact identity

$$2\beta(\mathcal{O}_B - \mathbb{E}\mathcal{O}_B) \stackrel{d}{=} g^\top Ag - \text{tr } A, \quad A = K^{-1/2}BK^{-1/2}. \quad (8.1)$$

Every bound above therefore applies at threshold $t = 2\beta h$ for an observable deviation h .

The same determinant is a free-energy difference:

$$\mathbb{E}e^{\theta\mathcal{O}_B} = \left[\frac{\det K}{\det(K - (\theta/\beta)B)} \right]^{1/2} \quad (8.2)$$

whenever the perturbed stiffness matrix is positive definite. The centering subtracts $\theta \text{tr } A/(2\beta)$ from its logarithm. Thus signed moment information controls an exact finite-volume free-energy response, as well as a probability tail.

When $B = K$, total energy is a scaled chi-square variable. More informatively, let B select the difference of energies in two orthogonal normal-mode sectors. Then A has eigenvalues $+1$ and -1 ; equal sector sizes give $s = 0$ and attain the signed envelope. The choice $\mathcal{O} = \mathcal{H}_+ - \frac{1}{2}\mathcal{H}_-$, with one eighth as many positive-sector modes, gives (7.1). The cancellation advantage has an immediate physical meaning: it concerns contrasts between fluctuating mode energies. These statements require a normalizable harmonic Gibbs measure; unconstrained zero modes must first be removed or confined.

8.2 Sharper guarantees for ordinary Gaussian trace estimates

For independent standard Gaussian probes $g_1, \dots, g_N$, set

$$\hat{\tau}_N = \frac{1}{N} \sum_{j=1}^N g_j^\top A g_j.$$

Its centered error is a quadratic form with matrix $\text{diag}(A/N, \dots, A/N)$. The parameters transform as

$$L_N = L/N, \quad v_N = v/N, \quad r_N = Nr, \quad s_N = s, \quad k_N = k.$$

For an absolute tolerance $\varepsilon > 0$, put $u = \varepsilon L/v$. Then

$$\mathbb{P}\{\hat{\tau}_N - \text{tr } A \geq \varepsilon\} \leq e^{-NrJ_R(s,k;u)}. \quad (8.3)$$

Together with reflection, a sufficient sample count for failure probability at most δ is

$$N \geq \frac{\log(2/\delta)}{r \min\{J_R(s, k; u), J_R(-s, k; u)\}}. \quad (8.4)$$

Replacing J_R by J gives the cubic-only guarantee.

For $s = 0$, the simplified coefficient guarantee improves the unrestricted sufficient sample count by the ratio $c(1)/c(0) = 0.81301005\dots$ before integer rounding, approximately 18.70% fewer probes. This comparison holds between the stated sufficient bounds; it is not an empirical performance claim or an information-theoretic lower bound for all trace-estimation algorithms. It also does not change the ε^{-2} small-error order of the ordinary Gaussian sample mean. Indefinite trace estimation and matrix-function applications are studied in [9]; more sophisticated methods such as Hutch++ address different query complexity goals [10].

8.3 Diagonal-free subgaussian forms

The Gaussian comparison used by Dadush et al. [4] also transfers our bounds to a broader class without decoupling losses.

Proposition 8.1. *Suppose $X_1, \dots, X_n$ are independent and $\mathbb{E}e^{hX_i} \leq e^{h^2/2}$ for every real h . If A is symmetric with zero diagonal, then every Chernoff upper-tail bound obtained from the Gaussian log-MGF envelopes above holds for $X^\top AX$, with the same matrix parameters.*

Proof. Conditional on all coordinates except i , the exponent $\theta X^\top AX$ is affine in X_i , because $A_{ii} = 0$. The defining subgaussian MGF inequality permits replacement of that coordinate by an independent standard Gaussian without decreasing the conditional exponential expectation. Replace coordinates one at a time, always retaining the original quadratic integrand. Tonelli's theorem handles the nonnegative integrand. Thus

$$\mathbb{E}e^{\theta X^\top AX} \leq \mathbb{E}e^{\theta g^\top Ag}.$$

The diagonal condition also gives $\text{tr } A = 0$, so the Gaussian right side is precisely the MGF of the centered Gaussian form used above. $\square$

For proxy parameter σ , meaning $\mathbb{E}e^{hX_i} \leq e^{\sigma^2 h^2/2}$, replace the threshold by t/σ^2 . No claim of sharpness for every subgaussian subclass is needed for this transfer.

8.4 Triangle-free graphs and random cuts

Let G be a simple triangle-free graph with $m > 0$ edges and adjacency matrix A . Closed-walk counting gives

$$\text{tr } A^2 = 2m, \quad \text{tr } A^3 = 0, \tag{8.5}$$

so $s = 0$. If X_i are independent uniform signs, the cut size is

$$\text{Cut} = \frac{m}{2} - \frac{1}{4}X^\top AX.$$

Applying Proposition 8.1 to either tail gives

$$\mathbb{P}\left\{\text{Cut} \leq \frac{m}{2} - d\right\} \leq \exp\left[-c(0) \min\left\{\frac{8d^2}{m}, \frac{4d}{L}\right\}\right], \quad d > 0, \tag{8.6}$$

and the same bound for $\text{Cut} \geq m/2 + d$. This uses vanishing cubic trace, which is weaker than spectral symmetry or bipartiteness. A fourth-moment refinement follows from

$$\text{tr } A^4 = 2m + 4 \sum_x \binom{\deg x}{2} + 8N_4, \tag{8.7}$$

where N_4 counts unoriented simple four-cycles. The terms count closed walks using one edge, two adjacent edges, or four distinct edges. For instance the five-cycle has $L = 2, v = 10, k = 3/4$ and Radau nodes $-3/4, 1$ with upper mass $3/7$.

A certified upper bound $L \geq \|A\|_{\text{op}}$ can replace the exact norm throughout the valid inequalities, after recomputing r, s, k . For graphs one may take the maximum degree. This can loosen the bound; claims of the optimal coefficient use the actual norm. Special graph families may admit stronger combinatorial estimates than (8.6); no optimal random-cut constant is asserted.

9 Computation and reproducible verification

9.1 What the included implementation computes

The module `code/spectral_concentration.py` computes the norm-only rate, signed rate and coefficient, Radau nodes, Radau rate and coefficient, directional and two-sided bounds, and a sufficient Gaussian-probe count. The input may be spectral moments or an explicitly supplied eigenvalue list. The formulas themselves do not require eigendecomposition once the moments are available.

If the complete spectrum is supplied, moment formation costs $O(n)$ arithmetic operations and each bound evaluation costs $O(1)$. For a general matrix accessible only by matrix-vector products, obtaining exact traces $\text{tr } A^3$ and $\text{tr } A^4$ may be costly. They are not free inputs. In particular, plugging unqualified noisy moment estimates into an extremal formula does not produce a certified probability inequality. A rigorous estimated-moment implementation must first construct a simultaneous confidence set and maximize over that set, accounting for its failure probability. That extension is a research problem below.

The numerical implementation uses `log1p` and convergent series near zero. At extreme tail sizes a floating-point saddle can round to a log-MGF singularity; the implementation chooses a representable point inside the domain. Any such point yields a valid Chernoff expression in exact arithmetic, although it need not be the exact optimizer. The implementation uses ordinary floating point and is not an outward-rounded interval proof.

9.2 Checks and exact artifacts

Run

```
python3 code/reproduce.py
```

from the package directory. It creates both figures and the CSV/JSON tables. The fixed seed is 20261008; no random samples of Q and no downloaded dataset are used. The audit includes:

- Five symbolic identities: the signed saddle derivative, the coefficient derivative identity, the reciprocal secant remainder, the defect antiderivative, and the Radau saddle polynomial.
- A deterministic set of 187 signed spectra, including projections, negative projections, balanced spectra, two-level spectra, zero eigenvalues, small eigenvalues, and random signed spectra.
- 1,309 comparisons of exact, Radau, cubic, and norm log-MGFs, and 1,122 comparisons of the resulting tail exponents.
- 45 independent scalar minimizations compared with the closed-form Radau optimizer, and 20 independent high-precision integrals for the spectral kernel.

- Nine exact rational tail certificates from (7.3), including a matrix of dimension 9,000. The standard-library certificate script also verifies (7.6) without rounding.

The hierarchy checks agree to floating-point roundoff; the detailed summary records the tolerances and maximum discrepancies. Boundary regressions cover invalid data and very large finite deviations. These computational checks support the derivations but are neither substitutes for the proofs nor a formal verification claim.

10 Interpretation, prior-art audit, and limits

10.1 What has been strengthened

The main theorem replaces one universal constant by the exact optimal function of a measurable spectral parameter. It completely characterizes the competition between small deviations and the quadratic/linear regime boundary. Its optimality is proved over actual finite-dimensional matrices at every exact real parameter, not only over formal Gamma laws.

The fourth-moment theorem gives a stronger full exponent with a closed optimizer. Its extremality is with respect to a specified moment relaxation, its equality cases are explicit, and its first omitted information has a nonnegative integral remainder. An actual matrix family attains the refinement and admits exact finite probability certificates. The gains are relevant to signed quadratic statistics, harmonic energy contrasts, and diagonal-free subgaussian forms.

10.2 What the literature review supports

The most relevant inspected sources were the original norm-only Gamma bound of Dadush et al., Gallagher et al.’s trace-based Gaussian concentration bounds, Moshksar’s September 2026 revision, Pang’s September 2026 Gamma comparison, and the classical Bai–Golub quadrature treatment [4, 5, 6, 7, 8]. Targeted searches used the exact proposed constants and the phrases signed cubic moment, skewness-dependent concentration, Gaussian quadratic form, and Gauss–Radau moment envelope. No matching theorem for the complete $c(s)$ frontier and its transition was located. Gallagher et al.’s discussion of asymmetry in the positive and negative spectrum is especially close in motivation.

An initial candidate was the unconditional coefficient $(1 - \log 2)/2$. The older literature ruled out a novelty claim for that coefficient. This matters because Moshksar’s recent discussion advertises smaller coefficients from particular inequality families. Those numbers are not a sufficient baseline for asserting a new global best constant. The comparisons in this article instead use the stronger established Gamma bound.

10.3 Limits of the result

The concentration problem is a central Gaussian problem, with a specified additional subgaussian extension for zero diagonal. No general noncentral or arbitrarily dependent subgaussian extension is proved. No asymptotic improvement in a matrix-query complexity exponent is claimed. The physics application concerns normalizable harmonic systems, and does not prove a new theorem about interacting critical spin models. The word optimal in the four-moment statement must retain its moment-relaxation qualifier.

Publication would require a broader literature check and external review of the proofs. These are limits on priority and validation, not assumptions left inside the mathematical arguments.

11 Further research questions and a proposed program

1. **Finite-matrix optimality with both moments fixed.** Determine the exact best coefficient for prescribed (r, s, k) when spectral multiplicities must be integers. The Radau envelope is a computable upper log-MGF bound, but its formal extremizer need not be a matrix. Start with two-node moment data and quantify the gap caused by multiplicity rounding. This is the most direct remaining mathematical question.
2. **A complete two-parameter sharp frontier.** For the relaxation, (5.10) gives an explicit function on $s^2 \leq k \leq 1$. Describe its level sets, its $1/4$ plateau, and the boundary where $J_R(s, k; 1) = 1/4$. Decide when its sharp coefficient is realized by actual matrix sequences with both normalized moments exact.
3. **Moment uncertainty with rigorous confidence guarantees.** Develop a robust optimization over simultaneous confidence intervals for v, p_3, p_4, L . The objective is to preserve a substantial part of the exponent gain while including the cost of estimating moments. Signed p_3 makes naive coordinatewise substitution unsafe. A useful deliverable would be a globally certified low-dimensional optimizer.
4. **Higher spectral-moment hierarchy.** Use right Gauss–Radau rules with more nodes to retain $p_2, \dots, p_{2m+2}$. Positivity of all spectral derivatives suggests a monotone sequence of log-MGF envelopes. Prove usable nonasymptotic convergence estimates that remain informative as the saddle approaches its pole; identify which additional moments provide the greatest improvement per unit computational work.
5. **Separate positive and negative spectral endpoints.** If $\lambda_{\max}$ and $-\lambda_{\min}$ differ markedly, the symmetric interval $[-L, L]$ is wasteful. The Hermite proof already works on a general interval. Determine the optimal coefficient surface after a normalization that treats the variance and both endpoints naturally, including all boundary regimes.
6. **Noncentral Gaussian quadratic forms.** The noncentral log-MGF has an additional rational spectral term depending on the mean vector. It is natural to introduce both variance-weighted and mean-weighted measures. The difficult part is a joint moment constraint, rather than applying two independent upper bounds that lose alignment information.
7. **Sharper prefactors for moment classes.** The present bounds optimize exponential rates. The exact example has a factor proportional to $r^{-1/2}$. Investigate uniform prefactor improvements under conditions controlling the tilted variance and the number of effective spectral modes. Such results would be substantially stronger for small failure probabilities.
8. **Graph structure beyond triangle counts.** For random cuts, p_3 and p_4 are short closed-walk counts. Combine the spectral envelope with combinatorial restrictions on these counts, and compare against specialized cut bounds. The zero-diagonal Gaussian comparison need not be sharp for random signs; identifying the additional slack could lead to a larger graph-specific constant.
9. **Certified numerical implementation.** Implement outward-rounded logarithms, square roots, and moment feasibility checks. Construct confidence bounds that remain valid under finite precision, and formalize the scalar interpolation and coefficient lemmas in Lean. The proof’s division into spectral integration, elementary calculus, and exponential tilting makes these independent verification targets.

10. **Weakly anharmonic systems.** For a Gibbs measure close to a harmonic reference, quantify the change of measure needed to transfer a signed-moment tail bound. Any such theorem must account explicitly for the interaction remainder and its temperature and volume dependence. The harmonic result supplies a reference inequality; it is not itself a justification for an interacting-model limit.

A A finite-rank certificate for the established endpoint constant

The unrestricted coefficient can be checked without invoking an asymptotic theorem as a black box. Let $Y \sim \chi_d^2$, $d \geq 1$, and $\varepsilon > 0$. Write $h(\varepsilon) = \varepsilon - \log(1 + \varepsilon)$. Then

$$\frac{e^{-dh(\varepsilon)/2}}{\varepsilon\sqrt{\pi d}} \frac{e^{-1/(6d)}}{1 + 2(1 + \varepsilon)/(d\varepsilon^2)} \leq \mathbb{P}\{Y \geq d(1 + \varepsilon)\} \leq \frac{e^{-dh(\varepsilon)/2}}{\varepsilon\sqrt{\pi d}}. \quad (\text{A.1})$$

These are elementary consequences of the Gamma integral and standard Stirling bounds [11]; they are included as a reproducible sharpness certificate, not as a new incomplete-Gamma estimate.

For completeness let $a = d/2$, $c = 1 + \varepsilon$, and $f(x) = x - 1 - \log x$. Substitution in the Gamma tail gives the integral $I = \int_c^\infty e^{-af(x)} dx/x$, multiplied by $a^a e^{-a}/\Gamma(a)$. Integration by parts yields

$$I = \frac{e^{-af(c)}}{a\varepsilon} - \frac{1}{a} \int_c^\infty \frac{x}{(x-1)^2} e^{-af(x)} \frac{dx}{x}.$$

The bracket is decreasing on $(1, \infty)$. If $M = e^{-af(c)}/(a\varepsilon)$, it follows that $M/[1 + c/(a\varepsilon^2)] \leq I \leq M$. The scaled Gamma function satisfies $1 < \Gamma^*(a) < e^{1/(12a)}$, which proves (A.1). At $\varepsilon = 1$ this becomes $e^{-dc_{\text{univ}}}/\sqrt{\pi d}$ times a factor tending to one. Any coefficient greater than c_{univ} must therefore fail in a sufficiently large finite dimension.

B Package contents and reproduction

The package contains the manuscript source and PDF, the principal Python module, a reproduction script, a standard-library exact certificate script, generated CSV and JSON data, vector and raster figures, build instructions, and a source/provenance record. The PDF is generated from this source with `pdflatex`. There are no external figure downloads and no runtime network requirement.

The exact-rational certificate script can run without scientific Python libraries. Reproducing figures and all audits requires NumPy, SciPy, Matplotlib, SymPy, and mpmath. Tested versions are recorded in the verification JSON. The user may integrate the source, results, and code into a research repository; no external repository was modified in preparing this package.

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